Analysis of Factors Influencing NCAAF AP Top 25 Rankings (2022 College Football Season)

Overview

This project analyzes what factors most strongly influenced the final AP Top 25 rankings in the 2022 college football season. The goal was to move beyond opinions and use data to understand what separates higher-ranked teams from the rest. This critical data that separates higher-ranked teams from the rest goes into important decisions like who will make the college football playoff.

From an analytics perspective, this is like evaluating performance drivers in any competitive environment, identifying which metrics truly matter when outcomes are determined.

Data & Approach

To answer this question, I combined two datasets:

- A dataset containing detailed team performance statistics (offense, defense, efficiency)
- The final AP Top 25 rankings and school's record for the 2022 season

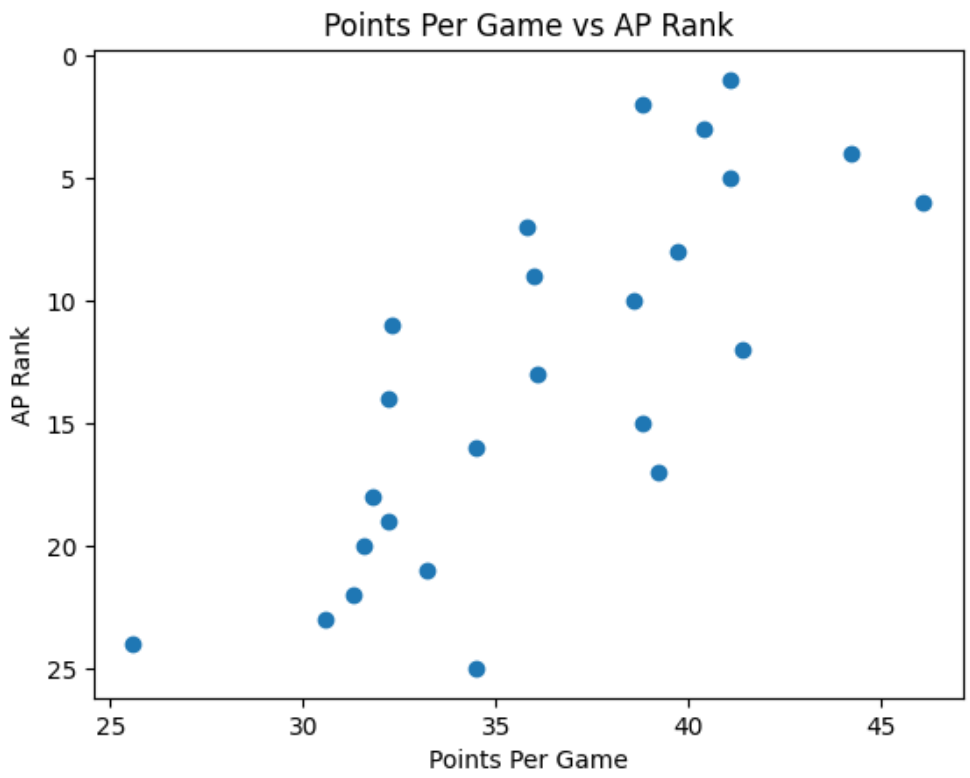
After cleaning and correcting team names, I merged the datasets so each team's performance metrics could be directly compared to its final ranking.

From there, I used correlation analysis to measure how strongly different performance metrics were associated with ranking. This allowed me to compare which factors had the most impact.

One important detail is that AP rankings are structured so that lower numbers represent better teams (Rank 1 is best). Because of this, stronger performance metrics appear as negative correlations, meaning better performance is associated with better (lower) rankings.

Key Findings

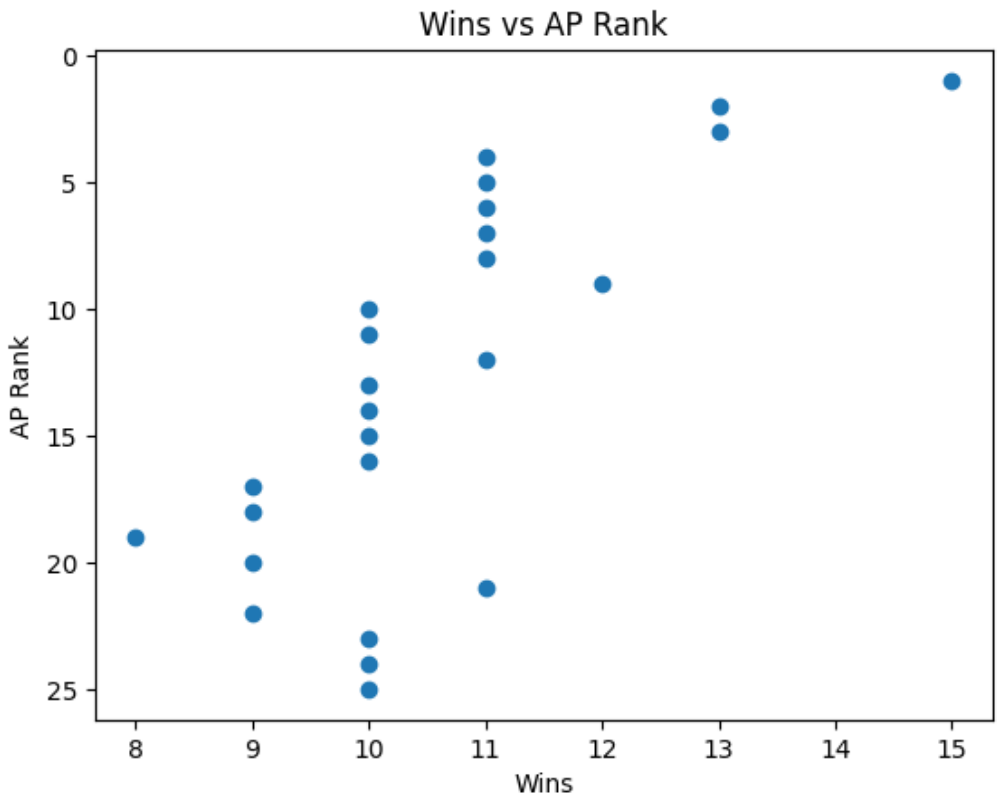
The results show that offensive performance and overall team success had the strongest relationship with ranking.



The relationship between points per game and AP ranking is clearly visible. Teams that scored more points per game are ranked higher.

The most impactful factors were:

- Points Per Game → strongest relationship with ranking
- Total Wins → nearly as strong
- Offensive efficiency metrics (yards per play, total yards per game)



Total wins also show a strong relationship with ranking. Teams with more wins consistently finished higher in the AP Top 25, reinforcing the importance of overall team success. In simple terms, teams that scored more gained more yards, and won more games were consistently ranked higher.

What About Defense?

One of the more interesting findings was that defensive performance had a much weaker relationship with ranking.

Metrics such as:

- Points Allowed per Game
- Yards Allowed per Game

showed relatively weak correlations compared to offensive stats.

This doesn't mean defense doesn't matter, but within the Top 25, it wasn't the primary factor separating teams.

Additional Insight: Top Teams vs the Rest

To take the analysis even further, I compared the Top 10 teams vs the remaining Top 25 teams.

The results reinforced the earlier findings:

- Top 10 teams scored significantly more points per game
- They gained more offensive yards
- They had stronger turnover margins
- They also won more games on average

This comparison made it clear that the highest-ranked teams didn't just perform slightly better, they were consistently stronger in key stats.

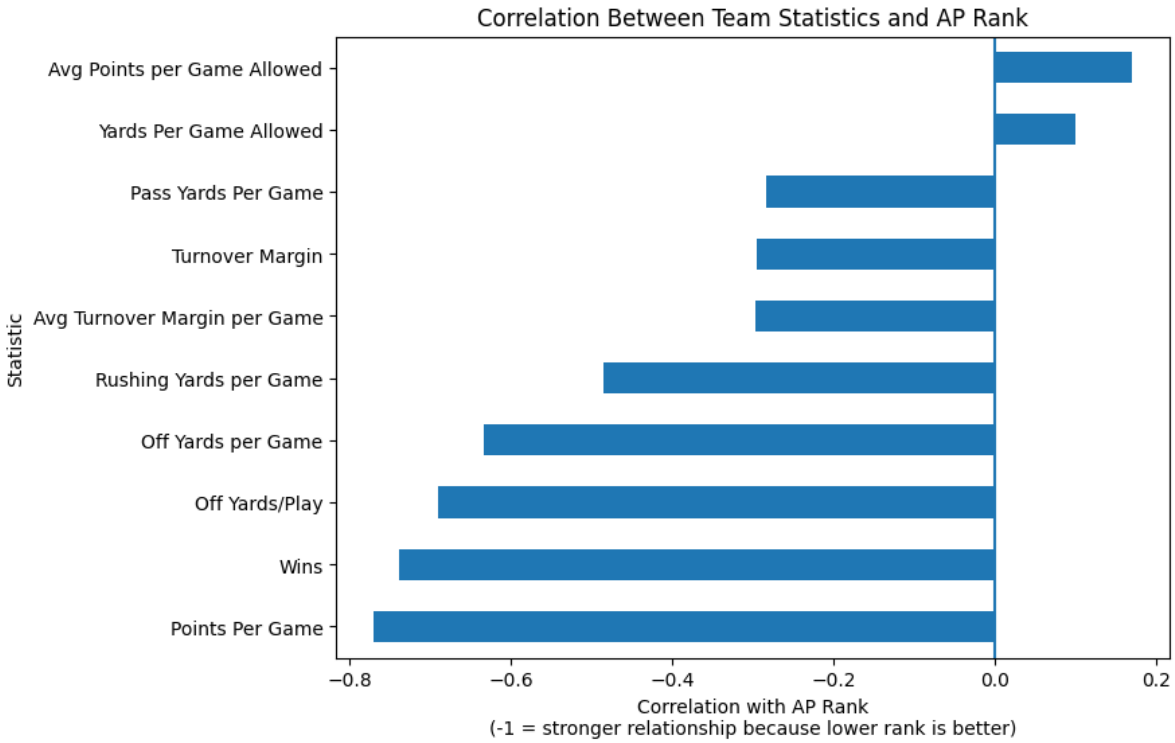
What This Means

From a broader perspective, this analysis suggests that AP rankings are driven more by visible offensive performance and results than by defensive metrics.

This makes sense when thinking about how rankings are determined:

- High-scoring teams standout more
- Wins are the most obvious measure of success
- Offensive dominance is easier to compare across teams

Statistical Results (Correlation Analysis)



The correlation chart highlights which statistics have the strongest relationship with AP ranking. Offensive metrics such as points per game and yards per play show the strongest relationships, while defensive statistics are noticeably weaker.

Conclusion

Overall, the data shows that offense and winning are the most important factors in determining AP rankings.

While defense and efficiency still play a role, they are less influential when comparing teams at the top level.

This project demonstrates how data analysis can be used to challenge assumptions and identify what truly drives outcomes, whether in sports or in other performance-driven environments.